

Q1. Tokenization Task

Given a paragraph, write a Python program (without external APIs) to:

- Split text into sentences
- Split sentences into words
- Count total tokens

Q2. Stopword Removal (Manual List)

Given a sentence and a predefined list of stopwords:

- Remove stopwords manually (without using online libraries)
- Display cleaned text

Q3. Stemming vs Lemmatization (Rule-based)

- Implement a simple rule-based stemmer (remove suffixes like *-ing*, *-ed*)

Q4. Morphological Analysis

Break the following words into morphemes and classify as:

- Free morpheme
- Bound morpheme

Words: *unhappiness*, *replayed*, *international*

Q5. Part-of-Speech Tagging (Rule-based)

Given a sentence:

- Assign POS tags using simple rules
(e.g., words ending in *-ly* → *adverb*)

Q6. Ambiguity Detection

Classify the following sentence as:

- Syntactic ambiguity
- Semantic ambiguity

Q7. Bag of Words Implementation

Given 3 sentences:

- Create vocabulary manually
- Construct Bag-of-Words matrix

Q8. TF-IDF Calculation (Manual)

For a given document:

- Compute Term Frequency (TF) of each word

- Display in tabular format

Q9. Cosine Similarity (Manual Calculation)

Given two vectors:

- Compute cosine similarity manually

Q11. Naive Bayes Classification (Numerical Problem)

Given:

- Word probabilities per class
- A test sentence

Q12. Hidden Markov Model (HMM) Basics

Given:

- Transition and emission probabilities

Q13. N-gram Text Generation

Write a Python program to:

- Generate text using bigram probabilities

Q14. BLEU Score

Given:

- Reference sentence
- Candidate sentence

Q15. Regex-based Text Cleaning

Write a Python program using regular expressions to:

- Remove punctuation
- Remove numbers
- Convert text to lowercase

Q16. Context-Free Grammar (CFG) Parsing

Given a simple grammar:

- Check whether a sentence is valid or not

Q17. Chunking (Rule-based)

Write rules to identify:

- Noun phrases (NP)
Example: (*Det* + *Adj* + *Noun*)

Q18. Using RNN perform sentiment analysis on movie reviews corpus of NLTK

Q19. Information extraction using named entity